

DBMS-lab(10)

S20230010045

1. View creation

```
mysql> CREATE VIEW branch_cust AS
-> SELECT account_relation.branch_name, depositor_relation.`customer-name`
-> FROM depositor_relation
-> JOIN account_relation ON depositor_relation.account_number = account_
relation.accountnumber;
Query OK, 0 rows affected (0.01 sec)
```

Trigger After insert into depositor

```
mysql>
mysql> CREATE TRIGGER depositor_relation_insert AFTER INSERT ON depositor_re
lation
-> FOR EACH ROW
-> BEGIN
->     INSERT INTO branch_cust (branch_name, `customer-name`)
->     SELECT account_relation.branch_name, NEW.`customer-name`
->     FROM account_relation
->     WHERE account_relation.accountnumber = NEW.account_number;
-> END //
Query OK, 0 rows affected (0.01 sec)
```

Trigger after insert into account

```
mysql> DELIMITER ;
mysql> select*
-> from depositor_relation_insert;
ERROR 1146 (42S02): Table 'database_name.depositor_relation_insert' doesn't
exist
mysql> DELIMITER //
mysql>
mysql> CREATE TRIGGER account_relation_insert AFTER INSERT ON account_relati
on
-> FOR EACH ROW
-> BEGIN
->     INSERT INTO branch_cust (branch_name, `customer-name`)
->     SELECT NEW.branch_name, depositor_relation.`customer-name`
->     FROM depositor_relation
->     WHERE depositor_relation.account_number = NEW.accountnumber;
-> END //
Query OK, 0 rows affected (0.01 sec)
```

Trigger after delete from depositor

```
mysql>
mysql> DELIMITER ;
mysql> DELIMITER //
mysql>
mysql> CREATE TRIGGER depositor_relation_delete AFTER DELETE ON depositor_re
lation
    -> FOR EACH ROW
    -> BEGIN
    ->     DELETE FROM branch_cust
    ->     WHERE `customer-name` = OLD.`customer-name`
    ->     AND account_number = OLD.account_number;
    -> END //
Query OK, 0 rows affected (0.01 sec)
```

Trigger after delete from account

```
mysql>
mysql> DELIMITER ;
mysql> DELIMITER //
mysql>
mysql> CREATE TRIGGER account_relation_delete AFTER DELETE ON account_relati
on
    -> FOR EACH ROW
    -> BEGIN
    ->     DELETE FROM branch_cust
    ->     WHERE branch_name = OLD.branch_name
    ->     AND account_number = OLD.accountnumber;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql>
```

2. Trigger on on delete

```
mysql>
mysql> CREATE TRIGGER after_account_delete
-> AFTER DELETE ON account_relation
-> FOR EACH ROW
-> BEGIN
->     DECLARE remaining_accounts INT;
->     DECLARE customer_name VARCHAR(255);
->     DECLARE done INT DEFAULT 0;
->
->     DECLARE owner_cursor CURSOR FOR
->         SELECT `customer-name`
->         FROM depositor_relation
->         WHERE account_number = OLD.accountnumber;
->
->     DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;
->
->     OPEN owner_cursor;
->
->     read_loop: LOOP
->         FETCH owner_cursor INTO customer_name;
->         IF done THEN
->             LEAVE read_loop;
->         END IF;
->
->         SELECT COUNT(*)
->         INTO remaining_accounts
->         FROM depositor_relation
->         WHERE `customer-name` = customer_name;
->
->         IF remaining_accounts = 0 THEN
->             DELETE FROM depositor_relation
->             WHERE `customer-name` = customer_name;
->         END IF;
->     END LOOP;
->
->     CLOSE owner_cursor;
-> END //
Query OK, 0 rows affected (0.01 sec)
```

3.

```
mysql>
mysql> CREATE PROCEDURE get_author_stats(IN author_name VARCHAR(255))
    -> BEGIN
    ->     DECLARE book_count INT DEFAULT 0;
    ->     DECLARE avg_price DECIMAL(10, 2);
    ->
    ->     SELECT COUNT(*) INTO book_count
    ->     FROM book
    ->     JOIN author ON book.author = author.author_id
    ->     WHERE author.authorName = author_name;
    ->
    ->     IF book_count > 0 THEN
    ->         SELECT AVG(price) INTO avg_price
    ->         FROM book
    ->         JOIN author ON book.author = author.author_id
    ->         WHERE author.authorName = author_name;
    ->     ELSE
    ->         SET avg_price = NULL;
    ->     END IF;
    ->
    ->     SELECT book_count AS Number_of_Books, avg_price AS Average_Price;

    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql> CALL get_author_stats('Kraig Muller');
+-----+-----+
| Number_of_Books | Average_Price |
+-----+-----+
|                2 |          650.00 |
+-----+-----+
1 row in set (0.01 sec)

Query OK, 0 rows affected (0.02 sec)
```